1. The problem is when I run the command “yaf -i bigflows.pcap” it gives an error telling me that it is refusing to write to the terminal. The reason is that by default, if no output option is given, it outputs IPFIX format which is not human readable format because it's in binary.

From: man yaf

```
Output Options
    These options control where yaf will send its output. yaf can write flows to an IPFIX file or export flows to an IPFIX collector
    over SCTP, TCP, UDP, or Spread.  By default, if no output options are given, yaf writes an IPFIX file to standard output.

    --out OUTPUT_SPECIFIER
```

From: ChatGPT

> The IPFIX file format itself is not a human-readable format, as it is a binary format designed for efficient storage and transmission of flow information. To interpret and analyze IPFIX files generated by YAF, you may use tools and libraries that support the IPFIX protocol or convert the binary IPFIX files to a more human-readable format using specialized software.

2. Application labeling can be used to identify the application protocol that is used in a network flow. Passive OS fingerprinting can be used to identify the operating system of a device that is sending network traffic. This information can be used to troubleshoot network problems, to identify malware and to trace attackers. Finally, deep packet inspection is when yaf inspects the contents of network packets in order to extract additional information. It can be used to identify malware,a dn to enforce security policies.
3. Yaf does application labeling by examining the packet payloads of network flows. It uses a many different combination of pattern matching and protocol-specific heuristics to identify the application protocol that is being used. The lines for bgp is “label 179 bgp” and for xmpp, it’s “label 5222 xmpp”.

Source: https://tools.netsa.cert.org/yaf2/applabel.html

4. Just seeing the flow as is, is kind of hard to determine the actual direction of the flow. However, the connection mostly seems to many clients trying to connect to many web servers.

```
2013-02-26 22:02:35.988 - 22:02:35.989 (0.001 sec) tcp 172.16.133.113:55582 => 172.16.139.250:5462 1ebe502f AF/ARF (3/120 ->)
2013-02-26 22:02:35.990 tcp 172.16.133.113:55582 => 172.16.139.250:5462 1ebe5030 AR/0 (1/40 ->)
2013-02-26 22:02:35.959 - 22:02:36.114 (0.155 sec) tcp 172.16.133.103:63406 => 216.115.222.200:443 70ef5af7:b1d60e07 S/AF:AS/AF (4/172 <-> 2/8
4) rtt 85 ms
2013-02-26 22:02:36.044 - 22:02:36.123 (0.079 sec) tcp 98.138.19.88:80 => 172.16.133.132:44296 b319e1c5:3d519ff1 AP/AF:AF/A (4/818 <-> 2/104)
rtt 0 ms
2013-02-26 22:02:36.080 - 22:02:36.234 (0.154 sec) tcp 172.16.133.132:38625 => 98.139.134.187:80 3ce3ed5d:d8c5786c S/APF:AS/APF (8/830 <-> 7/7
443) rtt 34 ms
2013-02-26 22:02:36.016 - 22:02:36.256 (0.240 sec) tcp 172.16.133.103:63407 => 66.151.158.187:443 5da77afe:88cffb99 S/AF:AS/AF (4/172 <-> 3/12
4) rtt 139 ms
2013-02-26 22:02:36.018 - 22:02:36.280 (0.262 sec) tcp 172.16.133.103:63408 => 216.219.112.250:443 f7a24030:adfd8781 S/AF:AS/AF (4/172 <-> 2/8
4) rtt 154 ms
2013-02-26 22:02:36.189 - 22:02:36.303 (0.114 sec) tcp 172.16.133.132:50395 => 76.13.6.174:80 3d2776ce:56398dbd S/APF:AS/APF (5/647 <-> 4/1156
) rtt 37 ms
2013-02-26 22:02:36.277 - 22:02:36.327 (0.050 sec) tcp 172.16.133.92:57607 => 216.219.115.250:443 17120721:c350d97a S/AF:AS/AF (4/172 <-> 2/84
) rtt 25 ms
2013-02-26 22:02:36.327 tcp 172.16.133.36:64948 => 208.85.40.50:80 59de96d2:2bf5bf86 AP/0:AR/0 (1/852 <-> 1/40) rtt 0 ms
2013-02-26 22:02:36.281 - 22:02:36.339 (0.058 sec) tcp 172.16.133.92:57608 => 67.217.66.244:443 afed8b47:46f1c329 S/AF:AS/AF (4/172 <-> 2/84)
rtt 25 ms
2013-02-26 22:02:36.249 - 22:02:36.361 (0.112 sec) tcp 172.16.133.132:54610 => 98.139.161.29:80 3d6af192:54e559ea S/APF:AS/APF (5/645 <-> 4/65
7) rtt 37 ms
2013-02-26 22:02:36.353 - 22:02:36.384 (0.031 sec) tcp 172.16.133.45:61190 => 173.70.53.92:41433 e6f58be7:acc6d952 AF/A:AF/A (2/80 <-> 2/80) r
tt 30 ms
2013-02-26 22:02:36.353 - 22:02:36.397 (0.044 sec) tcp 172.16.133.45:61200 => 24.125.152.21:51398 e2111084:72f146a5 AF/A:AF/A (2/80 <-> 2/80)
rtt 43 ms
2013-02-26 22:02:36.291 - 22:02:36.409 (0.118 sec) tcp 172.16.133.132:52323 => 98.139.240.23:80 3d067a87:00871536 S/APF:AS/APF (5/948 <-> 4/54
8) rtt 40 ms
2013-02-26 22:02:36.321 - 22:02:36.426 (0.105 sec) tcp 172.16.133.92:57609 => 68.64.30.250:443 6c04761d:e9ff1baf S/AF:AS/AF (4/172 <-> 2/84) r
:
```

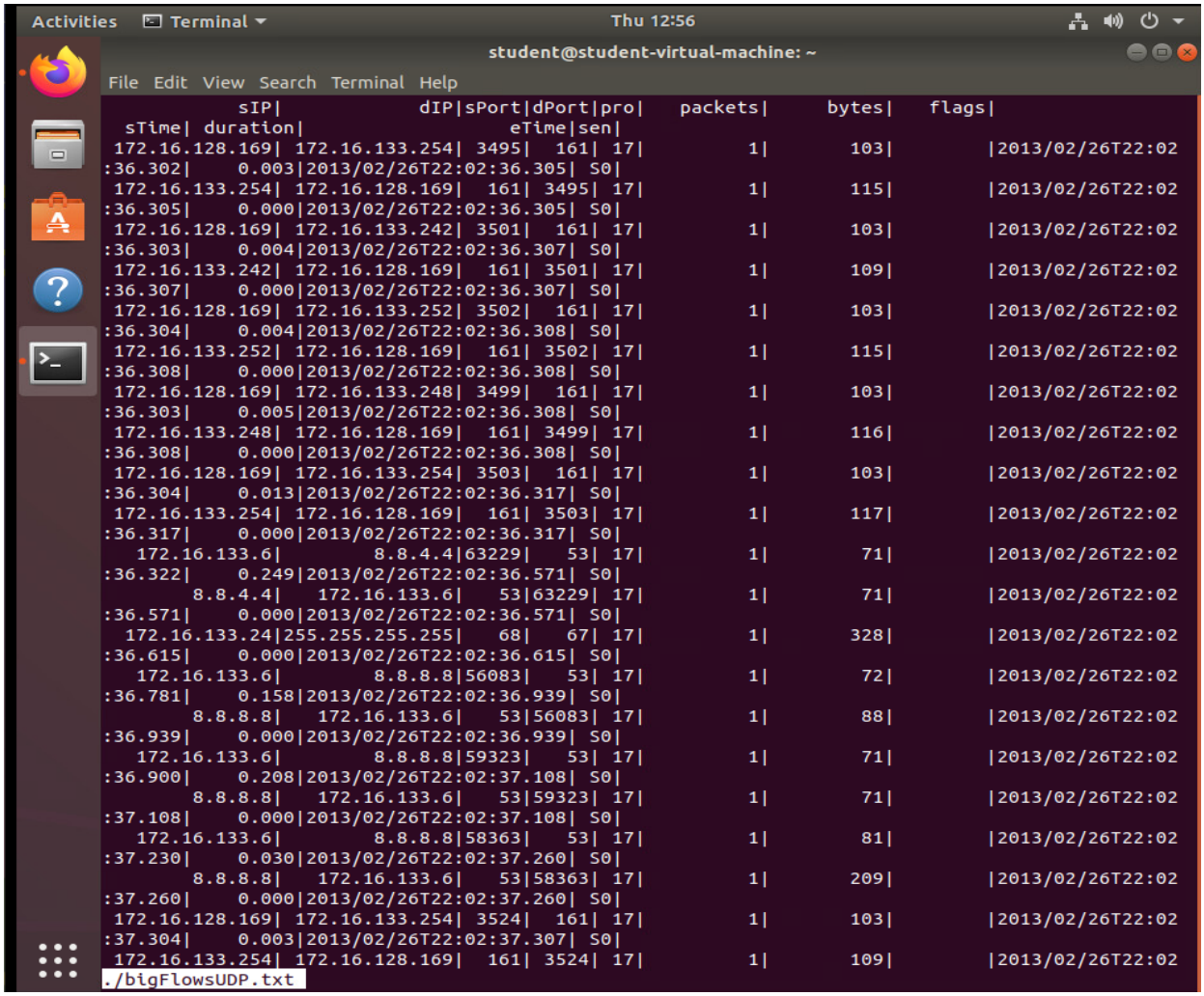
5. Here is a screenshot of the metadata

```
student@student-virtual-machine:~$ ls
bigFlows.pcap  bigFlows.rwf  bigFlows.yaf  bigFlows.yaf.txt  Desktop  Documents  Downloads  Music  outputyaf  Pictures  Public  Templates  Videos
student@student-virtual-machine:~$ rwfileinfo ./bigFlows.rwf
./bigFlows.rwf:
  format(id)          FT_RWGENERIC(0x16)
  version             16
  byte-order          littleEndian
  compression(id)     none(0)
  header-length       104
  record-length       52
  record-version      5
  silk-version        3.19.0
  count-records       41759
  file-size           2171572
  command-lines
                    1  rwipfix2silk --silk-output bigFlows.rwf ./bigFlows.yaf
student@student-virtual-machine:~$
```

6. The commands seem to organize 10 tcp traffic with the fields that are requested in the field option of command separating the fields with a line or '|'.

```
Activities   Terminal   Thu 12:46
student@student-virtual-machine: ~
File  Edit  View  Search  Terminal  Help
student@student-virtual-machine:~$ rwcut --num-rec=10 --fields=sip,dip,proto,sport,dport,stime
./bigFlows.rwf
            sIP|            dIP|pro|sPort|dPort|                 sTime|
 172.16.133.113| 172.16.139.250|  6|55582| 5462|2013/02/26T22:02:35.988|
 172.16.133.113| 172.16.139.250|  6|55582| 5462|2013/02/26T22:02:35.990|
 172.16.133.103|216.115.222.200|  6|63406|  443|2013/02/26T22:02:35.959|
216.115.222.200| 172.16.133.103|  6|  443|63406|2013/02/26T22:02:36.044|
   98.138.19.88| 172.16.133.132|  6|   80|44296|2013/02/26T22:02:36.044|
 172.16.133.132|   98.138.19.88|  6|44296|   80|2013/02/26T22:02:36.044|
 172.16.133.132| 98.139.134.187|  6|38625|   80|2013/02/26T22:02:36.080|
 98.139.134.187| 172.16.133.132|  6|   80|38625|2013/02/26T22:02:36.114|
 172.16.133.103| 66.151.158.187|  6|63407|  443|2013/02/26T22:02:36.016|
 66.151.158.187| 172.16.133.103|  6|  443|63407|2013/02/26T22:02:36.155|
student@student-virtual-machine:~$
```

7. Running the command piped it to rwcut, and then appended that to bigFlowsUDP.txt file. The screenshot below shows the beginning of the file using the less command



```
Activities   Terminal                    Thu 12:56
                          student@student-virtual-machine: ~
File  Edit  View  Search  Terminal  Help
            sIP|            dIP|sPort|dPort|pro|   packets|     bytes|   flags|
  sTime| duration|                    eTime|sen|
 172.16.128.169| 172.16.133.254| 3495|  161| 17|         1|       103|        |2013/02/26T22:02
:36.302|    0.003|2013/02/26T22:02:36.305| S0|
 172.16.133.254| 172.16.128.169|  161| 3495| 17|         1|       115|        |2013/02/26T22:02
:36.305|    0.000|2013/02/26T22:02:36.305| S0|
 172.16.128.169| 172.16.133.242| 3501|  161| 17|         1|       103|        |2013/02/26T22:02
:36.303|    0.004|2013/02/26T22:02:36.307| S0|
 172.16.133.242| 172.16.128.169|  161| 3501| 17|         1|       109|        |2013/02/26T22:02
:36.307|    0.000|2013/02/26T22:02:36.307| S0|
 172.16.128.169| 172.16.133.252| 3502|  161| 17|         1|       103|        |2013/02/26T22:02
:36.304|    0.004|2013/02/26T22:02:36.308| S0|
 172.16.133.252| 172.16.128.169|  161| 3502| 17|         1|       115|        |2013/02/26T22:02
:36.308|    0.000|2013/02/26T22:02:36.308| S0|
 172.16.128.169| 172.16.133.248| 3499|  161| 17|         1|       103|        |2013/02/26T22:02
:36.303|    0.005|2013/02/26T22:02:36.308| S0|
 172.16.133.248| 172.16.128.169|  161| 3499| 17|         1|       116|        |2013/02/26T22:02
:36.308|    0.000|2013/02/26T22:02:36.308| S0|
 172.16.128.169| 172.16.133.254| 3503|  161| 17|         1|       103|        |2013/02/26T22:02
:36.304|    0.013|2013/02/26T22:02:36.317| S0|
 172.16.133.254| 172.16.128.169|  161| 3503| 17|         1|       117|        |2013/02/26T22:02
:36.317|    0.000|2013/02/26T22:02:36.317| S0|
   172.16.133.6|        8.8.4.4|63229|   53| 17|         1|        71|        |2013/02/26T22:02
:36.322|    0.249|2013/02/26T22:02:36.571| S0|
        8.8.4.4|   172.16.133.6|   53|63229| 17|         1|        71|        |2013/02/26T22:02
:36.571|    0.000|2013/02/26T22:02:36.571| S0|
  172.16.133.24|255.255.255.255|   68|   67| 17|         1|       328|        |2013/02/26T22:02
:36.615|    0.000|2013/02/26T22:02:36.615| S0|
   172.16.133.6|        8.8.8.8|56083|   53| 17|         1|        72|        |2013/02/26T22:02
:36.781|    0.158|2013/02/26T22:02:36.939| S0|
        8.8.8.8|   172.16.133.6|   53|56083| 17|         1|        88|        |2013/02/26T22:02
:36.939|    0.000|2013/02/26T22:02:36.939| S0|
   172.16.133.6|        8.8.8.8|59323|   53| 17|         1|        71|        |2013/02/26T22:02
:36.900|    0.208|2013/02/26T22:02:37.108| S0|
        8.8.8.8|   172.16.133.6|   53|59323| 17|         1|        71|        |2013/02/26T22:02
:37.108|    0.000|2013/02/26T22:02:37.108| S0|
   172.16.133.6|        8.8.8.8|58363|   53| 17|         1|        81|        |2013/02/26T22:02
:37.230|    0.030|2013/02/26T22:02:37.260| S0|
        8.8.8.8|   172.16.133.6|   53|58363| 17|         1|       209|        |2013/02/26T22:02
:37.260|    0.000|2013/02/26T22:02:37.260| S0|
 172.16.128.169| 172.16.133.254| 3524|  161| 17|         1|       103|        |2013/02/26T22:02
:37.304|    0.003|2013/02/26T22:02:37.307| S0|
 172.16.133.254| 172.16.128.169|  161| 3524| 17|         1|       109|        |2013/02/26T22:02
./bigFlowsUDP.txt
```

8. The command rwuniq analyzes the data in the bigFlows.rwf file and generates unique records based on the specified fields and values. The field is proto, and value is records, packets, bytes. Finally the –sort-output field sorts out the output.
9. There are parts to the whole command. The first part “rwfilter –proto=6.17 –pass=stdout bigFlows.rwf” filters the flow records in bigFlows.rwf based on the specific protocols, which are TCP and UDP, and outputs the result to the standard output. The second part “rwuniq –field=sip –flows=50 –values=dip-distinct” takes the output from the first part and generates unique records based on the source IP. It also limits the output to 50 flows and considers distinct destination IPs. FInally the last part of the command “sort -nr -k 2,3, -t ‘|’” takes in the output from the second part and sorts the output numerically in reverse order based on the second and thirst columns where the ‘|’ is the field delimiter.
10. Silk provides support for IP sets and bags, which are specialized data structures for storing and manipulating IP addresses. IP sets are used to store collections of IP addresses, while IP bags are used to store mappings between IP addresses and values.

- rwsetbuild is a tool for creating IP sets. It takes an input file that contains a list of IP addresses, and it outputs an IP set file that can be used by other Silk tools.
- rwsettool is a tool for managing IP sets. It can be used to add, remove, and query IP sets.
- rwbagbuild is a tool for creating IP bags. It takes an input file that contains a list of IP address-value pairs, and it outputs an IP bag file that can be used by other Silk tools.
- rwbagtool is a tool for managing IP bags. It can be used to add, remove, and query IP bags.

Use Cases:

- IP sets and bags can be used for a variety of purposes, including:
- Access control: IP sets can be used to control access to resources, such as network services or files.
- Traffic filtering: IP sets can be used to filter network traffic, such as by blocking traffic from a particular range of IP addresses.
- Traffic analysis: IP sets and bags can be used to analyze network traffic, such as by identifying the most common source and destination IP addresses.
- Network security: IP sets and bags can be used to identify and track potential security threats, such as by identifying IP addresses that are associated with malicious activity.

Benefits:

The use of IP sets and bags provides several benefits, including:

Improved performance: IP sets and bags are optimized for storing and manipulating IP addresses, which can improve the performance of network applications.
Increased flexibility: IP sets and bags can be used to represent a wide variety of IP address relationships, which can make them more flexible than traditional data structures.
Enhanced security: IP sets and bags can be used to enforce security policies, which can help to protect networks from attack.

Overall, Silk's support for IP sets and bags provides a powerful and flexible way to store and manipulate IP addresses. This can be beneficial for a variety of network applications, including access control, traffic filtering, traffic analysis, and network security.